QUANTITATIVE STEREOLOGY FOR MICROSTRUCTURAL ANALYSIS

Ervin E. Underwood

Georgia Institute of Technology

Atlanta, Georgia 30332

Quantitative stereology, or metallography, is concerned with the quantitative characterization of microstructures in terms of their point, lineal, areal, and volume elements. By means of suitable two-dimensional measurements on the plane(s) of polish, statistically exact information can be obtained about the microstructural features in the three-dimensional space occupied by the alloy. These geometrical relationships have great generality and apply equally well to minerals, ceramics, plants, and metals.

In this review, we describe a unified system of symbols, the basic equations, and the more important measurements required. Special methods that provide additional information are given for the special class of materials that have oriented microstructures. Then, simple procedures for determining the "grain size" are presented, followed by a discussion of the measurement of particle characteristics. Finally, we summarize the important relationships applicable to projected images, such as obtained by thin foil transmission electron microscopy.

SYMBOLS

Table 1 shows the principal symbols used and gives examples of the combined notation in common usage. The term N_V, for example, refers to the number of objects per unit volume, and represents a fraction N/V_T, where N is the total number of objects in the arbitrarily selected test volume V_T. On the other hand, P_L (or P/L_T) stands for a measurement in which a test line of total length L_T is laid down repeatedly on the microstructure,

J. L. Mccall et al. (eds.), *Microstructural Analysis Tools and Techniques*
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TABLE 1. PRINCIPAL SYMBOLS AND COMBINED NOTATION

P, points of intersection, mm^0	
L, lines, mm	$P_L = P/L_T$
A, flat surfaces, mm^2	$L_A = L/A_T$
S, curved surfaces, mm^2	$A_A = A/A_T$
V, volumes, mm^3	$S_V = S/V_T$
N, number of objects, mm^0	$N_V = N/V_T$

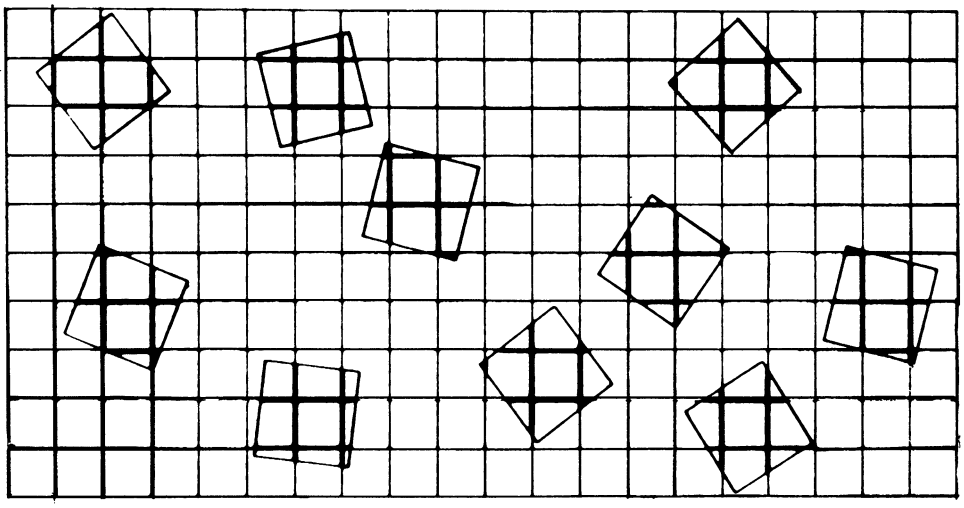


Figure 1. Equivalence of areal, lineal, and point ratios.

giving P number of point intersections with lineal features, perimeters, etc. Each symbol represents both a geometrical element as well as a specific dimension. Thus, the dimensionality of the combined terms and the equations is readily apparent and self-consistent.

BASIC EQUATIONS

The basic equations for points, lines, surfaces, and volumes in a microstructure are well known by now and are documented in numerous publications. (1-4) Consequently, derivations will not be given here. Instead, emphasis will be placed on the applications to actual microstructures.

Table 2 reveals in compact form the interrelationships between microstructural features that can be measured and those that can not. Arrows indicate those quantities that yield the normally inaccessible three-dimensional quantities V_V, S_V, I_V, and F_V. Below the triangular matrix are written the corresponding basic equations.

Studies of metallurgical systems and phenomena frequently require a knowledge of the volume fraction. The equations listed under Equation (1) state the equality of volume fraction to the areal ratio, lineal ratio, and point ratio of the selected phase as seen on random sections through the microstructure. The random squares sketched in Figure 1 show clearly the interrelationship of areal, lineal and point fraction measurements through the equations $A_A = L_L = P_P$. (6) The measurement of volume fraction is usually most efficiently performed by means of the point count, whereby the ratio P/P_T is obtained by the random placement of a point grid. Of course, under unusual circumstances, it might be preferable to obtain areal ratios with a planimeter or lineal ratios with a Hurlbut Counter. However, because of its speed, quantitative TV-scanning equipment (7) is now being used by industry for routine applications where large numbers of samples make automation economically feasible.

Equation (2) combines two important equations

$$S_V = 2\ P_L \tag{2a}$$

and

$$L_A = \frac{\pi}{2}\ P_L, \tag{2b}$$

both of which require the same P_L measurement. In those cases where the surface area per unit volume is required, we employ Equation (2a). For example, we might need the total grain boundary

TABLE 2. RELATIONSHIP OF MEASURED (○) AND CALCULATED (□) QUANTITIES

(Dimensions Arbitrarily Expressed in mm)

Microstructural Feature	Dimensions of Symbols			
	mm^0	mm^{-1}	mm^{-3} 2	mm^{-3}
Points	P_P	P_L	P_A	P_V
Lines	L_L	L_A	L_V	
Surfaces	A_A	S_V		
Volumes	V_V			

Basic Equations

$$V_V = A_A = L_L = P_P \qquad mm^0 \qquad (1)$$

$$S_V = \left(\frac{4}{\pi}\right) L_A = 2P_L \qquad mm^{-1} \qquad (2)$$

$$L_V = 2P_A \qquad mm^{-2} \qquad (3)$$

$$P_V = 2P_A P_L \qquad mm^{-3} \qquad (4)$$

surface area in a single-phase alloy. Also, S_V has been used in the analysis of precipitates in the grain boundary, for transformation products growing out of grain boundaries, and to obtain the interphase boundary area in a pearlitic steel.

When linear traces in the plane of polish are of interest, then L_A is the parameter of choice and we would use Equation (2b). In corrosion studies, for example, the length of the grain-boundary traces exposed to the corroding medium is the pertinent quantity. In other cases, we might arbitrarily consider the microstructural feature to be essentially linear. For thin twins, then, we would only count 1 for each twin intersected in a P_L traverse. For other twins, however, we might need to know the total twin-matrix interface length. We would then count the interfaces on each side of the twinned regions separately, and would count 2 each time the twin was intersected by the test line.

In the equation for line length per unit volume,

$$L_V = 2\ P_A\ , \tag{3}$$

we count the number of points in the test area that are caused by the intersections of the test plane with the linear elements of the microstructure. Equation (3) usually requires only one test plane for linear elements that are randomly oriented in space. However, randomness in the microstructure is not essential to Equation (3), since we can insure random sampling by selecting random orientations of the plane of polish.

Typical examples of linear elements in microstructures are dislocation lines, the grain edges where three adjacent grains come in contact, needle-like precipitate particles, and slag or oxide stringers. When dislocation etch pits are individually distinguishable, they can be counted (P_A) to get the dislocation density (L_V). Also, the "triple points" formed at the junctions of three grain-boundary traces are counted (P_A) to obtain the length per unit volume of grain-boundary edges (L_V).

The basic quantity P_V pertains to point elements per unit volume of the alloy. Unfortunately, such elements cannot be detected readily unless they have a finite size, or unless transmission electron microscopy is employed. For very small particles, the problem may actually be one of determining N_V, the number of particles per unit volume. In the special case of points generated by the intersections of superimposed systems of lines and surfaces, we have the equation (1)

$$P_V = 2\ P_A P_L\ , \tag{4}$$

where P_A and P_L are the number of points generated per unit area by test planes and per unit length by test lines, respectively.

ORIENTED STRUCTURES

Although the basic equations (Equations 1-4) apply to any kind of microstructure, special equations are available for special types of microstructures (1,3). For example, we frequently encounter oriented systems of lines or surfaces and may wish to determine the directional characteristics in addition to the mean values. In such cases, random sections are not employed, but, rather, sections that are transverse and/or longitudinal to a chosen direction.

Oriented systems can be classified conveniently into three types--namely, oriented lines in a plane, oriented lines in the (three-dimensional) alloy, and oriented surfaces in the alloy. These oriented systems can also be further described as completely oriented systems (in which all elements are parallel), or as partially oriented systems (in which both random and oriented elements occur). In both cases, however, there are clearly defined and recognizable orientation directions, and, in fact, there can be several orientation systems in the same microstructure, each with its own orientation direction or plane.

Examples of systems of oriented lines in a plane are afforded by twin traces in the plane of polish, or the grain-boundary traces in a heavily rolled sheet alloy. A system of partially oriented lines in a plane can be described by three equations. The equations refer to the random (ran) and oriented (or) portions of the system of lines, and to the total (tot) length of line per unit area. They are

$$(L_A)_{ran} = \frac{\pi}{2} (P_L)_{||} \tag{5a}$$

$$(L_A)_{or} = (P_L)_{\perp} - (P_L)_{||} \tag{5b}$$

$$(L_A)_{tot} = (P_L)_{\perp} + 0.571\ (P_L)_{||} \tag{5c}$$

in which the $(P_L)_{\perp}$ and $(P_L)_{||}$ measurements are made with test lines placed perpendicular and parallel to the orientation direction, respectively. Note that $(L_A)_{tot} = (L_A)_{ran} + (L_A)_{or}$. Of course, if the system of lines is completely oriented, $(P_L)_{||}$ will be zero and $(L_A)_{tot} = (L_A)_{or} = (P_L)_{\perp}$.

There are many common microstructures with features that can be described as a system of oriented lines in space (i.e., lines

that are distributed throughout the alloy). Examples of such systems are elongated nonmetallic inclusions, the parallel rods in unidirectionally solidified eutectics, and the oriented dislocation arrays, all of which show pronounced linear directional characteristics. For a partially oriented system of lines in the alloy, we have

$$(L_V)_{ran} = 2\ (P_A)_{||} \tag{6a}$$

$$(L_V)_{or} = (P_A)_{\perp} - (P_A)_{||} \tag{6b}$$

$$(L_V)_{tot} = (P_A)_{\perp} - (P_A)_{||} \tag{6c}$$

where the subscripts have the same meanings as before, and $(P_A)_{\perp}$ and $(P_A)_{||}$ refer to measurements of point density on planes perpendicular and parallel to the orientation direction, respectively. If the system of lines is completely oriented, $(P_A)_{||}$ is zero and $(I_V)_{tot} = (L_V)_{or} = (P_A)_{\perp}$. Although some microstructural features are not truly linear, they can be considered so for practical purposes if they have sufficient linearity. Of course, if the cross-sectional thickness is too great, there will be difficulty in making the $(P_A)_{||}$ measurements.

The third major type of oriented structure consists of surfaces in the alloy. Examples of oriented planar features in the microstructure are pearlite in steel and the lamellae in unidirectionally solidified eutectics. These types of oriented surfaces belong to the subclassification known as "planar" orientation, since the planar surfaces are essentially parallel to an orientation plane (or planes). The three applicable equations for a partially oriented system of surfaces with planar orientation are expressed as

$$(S_V)_{ran} = 2\ (P_L)_{||} \tag{7a}$$

$$(S_V)_{or} = (P_L)_{\perp} - (P_L)_{||} \tag{7b}$$

$$(S_V)_{tot} = (P_L)_{\perp} + (P_L)_{||} \tag{7c}$$

where the subscripts have the same meanings as before, and $(P_L)_{\perp}$ and $(P_L)_{||}$ are measurements made perpendicular and parallel to the orientation plane, respectively. If the system of surfaces is completely oriented, $(P_L)_{||}$ is zero and $(S_V)_{tot} = (S_V)_{or} = (P_L)_{\perp}$. An example of the type of calculation that follows the above analysis for a partially-oriented system of surfaces with planar orientation is given in Reference 1.

The results of measurements based on Equations 5, 6, and 7 can be expressed conveniently as a fraction, whereby the oriented portion of the partially oriented system is divided by the total amount. The resulting fraction gives an easily recognized, natural scale of anisotropy varying from 0 for no anisotropy to unity for a completely oriented system. This ratio is called the degree of orientation (3), Ω, and is defined as

$$\Omega_{12} = \frac{(L_A)_{or}}{(L_A)_{tot}} = \frac{(P_L)_{\perp} - (P_L)_{||}}{(P_L)_{\perp} + 0.571\ (P_L)_{||}} \tag{8a}$$

$$\Omega_{13} = \frac{(L_V)_{or}}{(L_V)_{tot}} = \frac{(P_A)_{\perp} - (P_A)_{||}}{(P_A)_{\perp} + (P_A)_{||}} \tag{8b}$$

$$\Omega_{34} = \frac{(S_V)_{or}}{(S_V)_{tot}} = \frac{(P_L)_{\perp} - (P_L)_{||}}{(P_L)_{\perp} + (P_L)_{||}} \tag{8c}$$

where the subscripts 12 refer to lines in a plane, 13 to lines in a volume, and 23 to surfaces in a volume (1).

GRAIN SIZE

Grain sizes, or "diameters", have been determined by many methods in the past. However, because the grains normally found in alloys have irregular shapes, the definition of a diameter is usually quite arbitrary. In fact, only for a sphere is the diameter an unambiguous concept.

Fortunately, there is a quantitative length parameter of great generality that gives a unique, assumption-free value for any granular space-filling structure, regardless of grain shape, size, or position. This "diameter" is the mean intercept length $\bar{L}_3$ obtained from simple L_2 intercept measurements on the plane of polish. For many random planes, of course, the averaged $\bar{L}_2$ values become the true three-dimensional $\bar{L}_3$ parameter.

For space-filling grains, the mean intercept length is defined as

$$\bar{L}_3 = \frac{1}{N_L} = \frac{L_T}{P_M} \tag{9}$$

where N_L is the number of interceptions of grains per unit length of test line, L_T is the total test line length, P is the number of grain boundary intersections, and M is the magnification employed. Note that N = P, or equally, that $N_L = P_L$.

The physical nature of the mean intercept length is as follows. If we imagine an irregularly shaped convex grain through which lines are passed at all possible locations and angles, the intercept lengths will vary from zero to a maximum at a particular location and direction. The mean value of the intercept lengths will lie somewhere between these two extremes. In a sphere, for example, the mean intercept length is equal to two-thirds of the diameter.

Moreover, even for a concave grain, there is no difficulty, since we define N_L as the number of interceptions of grains per unit length of test line. This definition allows for the possibility that a particle with an irregular outline may be intercepted more than once by the same test line. Thus, we maintain the relationship $P_L = N_L$ for space-filling grains, regardless of their shapes. (See Tables 3 and 4, first column.)

Using the mean intercept length as described above, then it can be thought of as a lineal measure of the "size", even in a system of irregularly shaped volume elements. No assumptions as to size or sphericity or other shapes are required.

The mean intercept length is also directly related to S_V, the grain boundary surface area per unit volume. Since for space-filling cells or grains, $\bar{L}_3 = 1/N_L$ and $N_L = P_L$, we obtain

$$S_V = 2\ P_L = 2\ N_L = 2/\bar{L}_3 \tag{10}$$

It should be recalled that S_V also has a unique value for any system of surfaces in a volume.

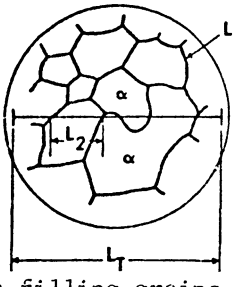
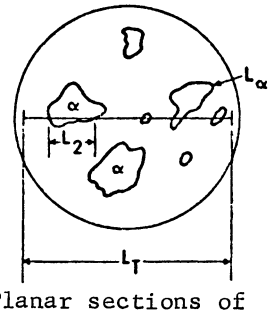
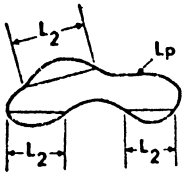
When $\bar{L}_3$ is expressed in millimeters, it gives the same value as the Intercept (or Heyn) Procedure described in ASTM E112-63 "Standard Methods for Estimating the Average Grain Size of Metals." This article also gives the basis for the ASTM grain size number "N", which is defined by the equation

$$n = 2^{N-1} \tag{11}$$

where n is the number of grains per square inch at 100X. (Note that $n = N_A$ in the notation of this article.) Normally to obtain the ASTM grain size number, one must count a minimum of at least 50 grains in each of three areas, determine the number per square inch, and convert this value to the equivalent one at 100X. Then, substitution in Equation 11, or recourse to tables, gives "N".

Hilliard has devised a particularly quick and useful method for determining an equivalent "N" by means of the P_L count. (8) He provides two circular test figures of known lengths as depicted in Figure 2 (not shown to size). The test circles can be

TABLE 3. TWO-DIMENSIONAL PLANAR FIGURES

$A_A=1$	$(A_A)_\alpha<1$	Area = A
		
Area-filling grains of one phase	Planar sections of α-phase in a matrix	Isolated single figure
$N_L = 1/2 + 4 + 1/2 = 5$	$N_L = 4$	$N_L = 1,2,\ldots$
$P_L = 5$	$P_L = 8$	$P_L = 2,4,\ldots$
$P_L = N_L = 1/\bar{L}_2$	$P_L = 2N_L = 2\,(A_A)_\alpha\,/\bar{L}_2$	$P_L = 2N_L$
Saltykov:		
$L_A = \frac{\pi}{2} P_L = \frac{\pi}{2} N_L$	$(L_A)_\alpha = \frac{\pi}{2} P_L = \pi\, N_L$	$\frac{L_p}{A} = \frac{\pi}{2}\frac{P_L}{P_P}$
Tomkeieff:		
$\bar{L}_2 = \frac{\pi \bar{A}}{2\bar{L}_p} = \frac{\pi}{2L_A}$	$\bar{L}_2 = \frac{\pi \bar{A}_\alpha}{(\bar{L}_p)_\alpha} = \frac{\pi (A_A)_\alpha}{(L_A)_\alpha}$	$\bar{L}_2 = \frac{\pi A}{L_p}$
Chalkley:		
$\bar{L}_2 = \frac{\ell h}{2P}$	$\bar{L}_2 = \frac{\ell h}{P} = \frac{\pi \bar{A}_\alpha}{(\bar{L}_p)_\alpha}$	$\bar{L}_2 = \frac{\ell h}{P} = \frac{\pi A}{L_p}$

$\bar{L}_p$ = Mean perimeter length per planar figure = L_A/N_A.

$\bar{L}_2$ = Mean intercept length of planar figures = $L_L/N_L = A_A/N_L$.

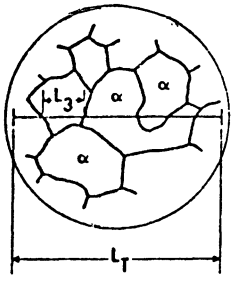
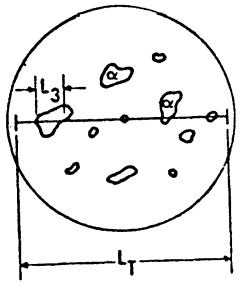
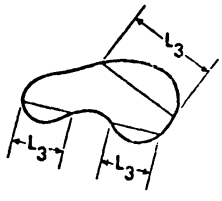
ℓ = Constant length of short test lines, thrown randomly on microstructure.

h = Number of (end) points of ℓ-lines that hit in area of interest.

P = Number of intersections with perimeters made by ℓ-lines.

L_T = Length of test line.

TABLE 4. THREE DIMENSIONAL GRAINS AND PARTICLES

$V_V = 1$	$(V_V)_\alpha < 1$	Volume = V
		
Space-filling grains of one-phase	Dispersed particles of α-phase in a matrix	Isolated single particle
$N_L = 1/2 + 4 + 1/2 = 5$	$N_L = 4$	$N_L = 1, 2, \ldots$
$P_L = 5$	$P_L = 8$	$P_L = 2, 4, \ldots$
$P_L = N_L = 1/\bar{L}_3$	$P_L = 2N_L = 2\ (V_V)_\alpha/\bar{L}_3$	$P_L = 2N_L$
Saltykov:		
$S_V = 2P_L = 2N_L$	$(S_V)_\alpha = 2P_L = 4N_L$	$\frac{S}{V} = \frac{2P_L}{P_P}$
Tomkeieff:		
$\bar{L}_3 = 2\,\frac{\bar{V}}{\bar{S}} = \frac{2}{S_V}$	$\bar{L}_3 = \frac{4\bar{V}_\alpha}{S_\alpha} = \frac{4(V_V)_\alpha}{(S_V)_\alpha}$	$\bar{L}_3 = 4\frac{V}{S}$
Chalkley:		
$\bar{L}_3 = \frac{\ell h}{2P}$	$\bar{L}_3 = \frac{\ell h}{P} = \frac{4\bar{V}_\alpha}{\bar{S}_\alpha}$	$\bar{L}_3 = \frac{\ell h}{P} = \frac{4V}{S}$

$\bar{S}_\alpha$ = Mean surface area of α-particles = $(S_V)_\alpha / N_V$.

$\bar{L}_3$ = Mean intercept length of three-dimensional bodies = $L_L/N_L = A_A/N_L = V_V/N_L$.

ℓ = Constant length of short test lines, thrown randomly on microstructure.

h = Number of (end) points of ℓ-lines that hit in phase of interest.

P = Number of intersections with surfaces made by ℓ-lines.

L_T = Length of test line.

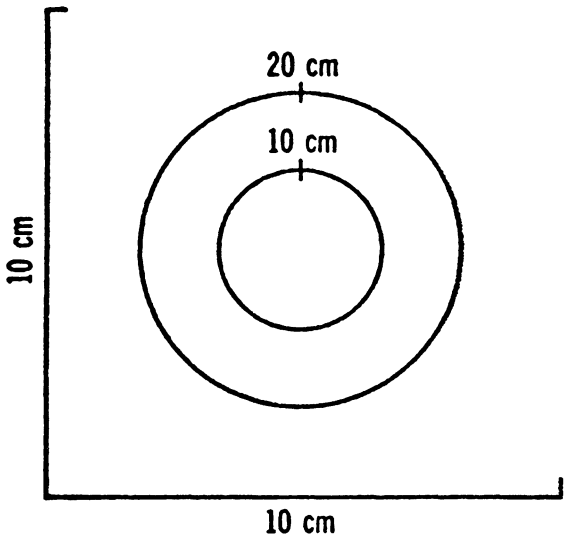


Figure 2. Circular test figures for grain size measurements.

reproduced on plastic sheet (for analyzing photomicrographs) or placed on the ground glass screen of a metallograph. The best method, however, is to use the test circle as a reticule in the focusing eyepiece of a bench microscope.

The operator selects one of the circles and a magnification for his sample which will result in more than six intersections per application of the circle, on the average. For equiaxed grains that do not vary much in size, the circle is applied to the microstructure until about 35 intersections are obtained. This number will insure that a standard deviation of 0.3 unit in G, the equivalent grain size number, is attained.

In order to calculate G, we use Hilliard's equation, which can be written as

$$G = -10.00 - 6.64 \log \bar{L}_3(\text{cm}) \qquad (12)$$

with $\bar{L}_3 = L_T/PM$ as before. Here P is the total number of grain-boundary intersections made by a test circle laid down several times to give a total length L_T (in cm) on a field viewed at any magnification M. To demonstrate the operation of Equation 12, suppose that the 10 cm circle is applied four times to a microstructure at 250X, giving a total of 36 intersections. G then

equals -10-6.4 log (40/36 x 250) or 5.6. Thus the equivalent grain size number is obtained directly, quickly, and with maximum efficiency, since no more intersections are counted than those required to insure the desired accuracy. A nomograph is also provided (8) for the graphical solution of Equation 12 and is reproduced in Figure 3.

PARTICLE RELATIONSHIPS

Many of the relationships pertaining to particulate structures apply with equal validity to second-phase regions, voids, boundary precipitates, etc. One of the more important of the general relationships involves the mean free distance, λ, which is the mean edge-to-edge distance, along random straight lines, between all possible pairs of particles. For α-phase particles, the mean free distance is

$$\lambda = \frac{1-(V_V)_\alpha}{N_L} \tag{13}$$

where $(V_V)_\alpha$ is the volume fraction of the α particles and N_L is the number of particle interceptions per unit length of test line. Equation (13) is valid regardless of size, shape, or distribution of the particles and represents a truly three-dimensional interparticle distance. This parameter is of great importance in studies of strength or other mechanical properties and has been used in several different ways.

There are also other types of interparticle distance and spacing parameters, such as the mean particle spacing, σ, which is essentially the mean particle center-to-center length. The defining equation for mean particle spacing is simply

$$\sigma = \frac{1}{N_L} \tag{14}$$

where here N_L is the number of particle interceptions per unit length of random test lines. The parameter σ is characterized by the great ease with which it can be measured, since only a simple particle intersection count is needed. It is also related to λ through the mean intercept length, $\bar{L}_3$, by the equation

$$\lambda = \sigma - (\bar{L}_3)_\alpha \tag{15}$$

where $(\bar{L}_3)_\alpha$ for particles of α phase is defined by $(\bar{L}_3)_\alpha = (\bar{L}_L)_\alpha / N_L$. This is a completely general and assumption-free expression, valid for particles of any size or configuration.

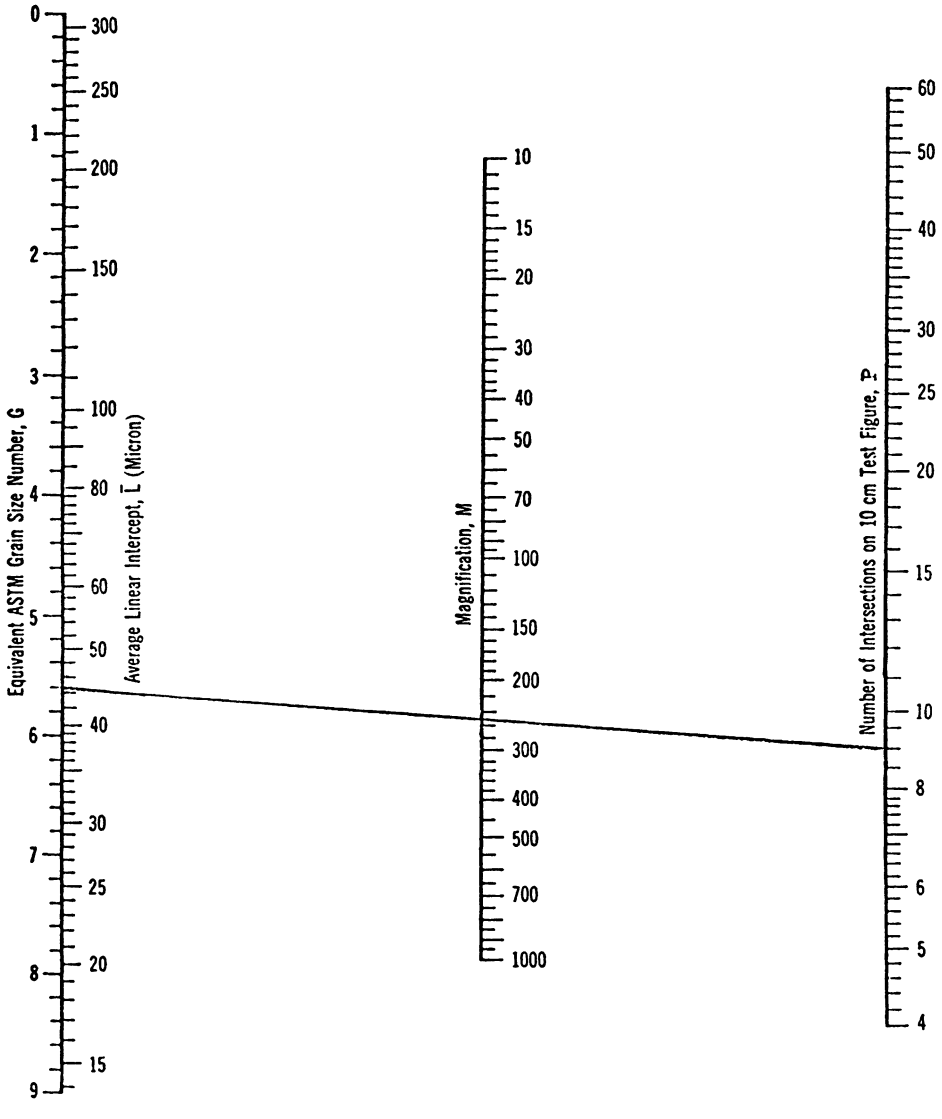


Figure 3. Nomograph for obtaining grain size numbers.

The mean particle intercept length, $(\bar{L}_3)_\alpha$, is a companion term to λ, in that λ is the mean matrix intercept distance and $(\bar{L}_3)_\alpha$ is the mean particle intercept distance. They are related through the expression for a two-phase or particulate structure of α phase by

$$\lambda = (\bar{L}_3)_\alpha \left[\frac{1-(V_V)_\alpha}{(V_V)_\alpha} \right] \tag{16}$$

where λ is the mean free distance between particles that have a volume fraction $(V_V)_\alpha$ and mean intercept length $(\bar{L}_3)_\alpha$. Equation (16) has been used to verify the value of volume fraction in a two-phase alloy in which the size and configuration of the dark second phase could be varied readily by heat treatment. However, the volume fraction remained relatively constant. Thus the (constant) volume fraction obtained from the slope of the λ versus $(\bar{L}_3)_\alpha$ curve (73.2 vol %) corresponded well with the volume fraction determined by point counting (73.5 volume %) and from chemical analysis (71.4 vol %).

It should be noted that the mean intercept lengths for space-filling grains and for particles are related through the general expression

$$\bar{L}_3 = \frac{L_L}{N_L} \quad . \tag{17}$$

In single-phase alloys, L_L (or V_V) = 1 and Equation 9 is obtained. Otherwise, for two-phase or particulate alloys, L_L (or V_V) has a value less than one and Equation (17) is used. Also $2N_L = P_L$ applies for particles instead of $N_L = P_L$ for the single-phase alloys.

An example of the application of the mean intercept lengths is seen in the well-known relation,

$$R = \frac{4r}{3\ V_V} \quad , \tag{18}$$

where R is the mean grain radius and r the mean particle radius. Experimentally, $\bar{L}_3$ and $(\bar{L}_3)_\alpha$ can be obtained and used for the grain "diameter" and particle "diameter", respectively, giving good agreement between calculated and measured grain sizes.

From the foregoing discussion of grain and particle characteristics, it can be seen that there are many points of similarity in their geometrical properties. On the plane of polish the grain boundaries and particle interphase traces are both measured by L_A or L_p (the perimeter length); the intercept distances for both

grains and particles are expressed by $\overline{L}_2$ or $\overline{L}_3$; and the surface area per particle or grain, S_α/V_α; and the surface area per unit volume of sample, S_V, apply equally to both volume elements.

However, there is one distinctive difference between grains and particles. Since the grains are space filling, all grain boundaries are shared by two contiguous grain faces. Particles, on the other hand, do not usually occupy 100 percent of the alloy, thus sharing of particle interfaces does not occur as much. In order to emphasize these differences, Table 3 summarizes the pertinent equations for planar figures, both area-filling and separated, while Table 4 does the same for grains and particles. In general, the quantities in the second and third columns are double those in the first column. It should also be noted that the parameters defined in Tables 3 and 4 apply equally as well to interpenetrating two-phase structures as to simple particulate systems.

PARTICLE-SIZE DISTRIBUTIONS

Several methods are available for obtaining the spatial size distribution of spheres from the size distribution of their planar sections. Procedures are also available for convex particles of arbitrary shape (9), ellipsoids (10), pentagonal dodecahedrons (11), a statistical grain shape (12), and the spacings in lamellar structures (13). Whereas the equations for the simpler particles are capable of statistically exact solutions, this is not the case for size distributions of real particles with irregular shapes. Consequently, assumptions are required, with a corresponding loss in the accuracy of the results.

The three main types of quantities measured on planar sections are the section diameters, section areas, or section chords. (14) From the resulting two-dimensional size distribution, we can calculate the true spatial size distribution or particles or grain sizes.

Frequently, however, we do not need the size distribution curve in order to characterize a microstructure. In fact, in order to relate the size distribution to some material property, numerical parameters are required instead of the curve itself. Generally, all we need to represent a size-distribution curve is the mean diameter $\overline{D}$, the standard deviation $\sigma(D)$, and the total number of particles per unit volume N_V. These parameters can be obtained from the analysis of the particle size distribution, or in some cases, directly from the appropriate experimental data.

A comparison is made in Table 5 of selected methods for obtaining the spatial size distribution of systems of particles with specific shapes. Methods that deal with nonspherical particles are noted, as are those that employ nonanalytical solutions. Two methods, characterized by unusual simplicity, are given by Saltykov (9) and Bockstiegel (15). We will discuss briefly the procedures involved in the calculations of size distributions by these two authors.

Saltykov's method (9) is based on relative section areas, A/A_{max}, from the planar distribution curve of sections through a sphere and is equally applicable to any system of convex particles of one shape. A logarithmic scale of diameters is used with the factor $10^{-0.1} = 0.7943$. Thus, for sectional areas, the factor is $(10^{-0.1})^2 = 0.6310$. Table 6 lists group numbers, the corresponding diameters, and the relative section area limits required for the class intervals.

Since the section area is specified in terms of the largest section area, it is important that a large number of sections be examined in order to get the correct value of A_{max}. Next, we count the number of sections per unit area, $(N_A)_i$, and group them according to the area limits specified in Table 6. A series of graded circles serves this purpose quite well. These values are then substituted in Saltykov's working equation, which has precalculated coefficients and provisions for up to 12 class intervals. The equation is

$$(N_V)_j = \frac{1}{D_j} \Big[1.6461(N_A)_i - 0.4561(N_A)_{i-1} - 0.1162(N_A)_{i-2} - 0.0415(N_A)_{i-3} - 0.0173(N_A)_{i-4} - 0.0079(N_A)_{i-5} - 0.0038(N_A)_{i-6} - 0.0018(N_A)_{i-7} - 0.0010(N_A)_{i-8} - 0.003(N_A)_{i-9} - 0.0002(N_A)_{i-10} - 0.0002(N_A)_{i-11} \Big] . \quad (19)$$

$(N_V)_j$ represents the number of particles per unit volume in the j-th class interval, and j is an integer with any value from 1 to 12. The largest particle size corresponds to a value of j = 1. The i-values for the sections depend on the particular sphere size, or j-value, chosen for calculation. Thus, as each value of j is selected, i is set equal to j, and this determines the number of terms used inside the square brackets. For example, to calculate the value of $(N_V)_5$, we would only use the first five terms

TABLE 5. COMPARISON OF METHODS FOR OBTAINING SIZE DISTRIBUTIONS

Method	Particle Shape	Characteristics of Method*	Remarks
DIAMETERS:			
DeHoff (10)	Ellipsoids	T,I	Uses axial ratios. Shape factors obtained from curve.
Scheil and Wurst (12)	Statistical shape	T,S	Based on ingot iron grain.
Schwartz-Saltykov (1)	Spheres	T,I	
Paulus (11)	Pentagonal dodecahedrons	T,C,L.S	Method based on d/d max distribution curve.
AREAS:			
Saltykov (9)	Spheres, convex particles	WE,I,L	Method based on A/A max distribution curve.
CHORDS:			
Lord and Willis (1)	Spheres	G,I	
Cahn and Fullman (13)	Lamellar structures	G,I	Slopes taken from experimental distribution curve.
Bockstiegel (15)	Spheres	WE,I,L	No coefficients required simplified version.

*Explanation of Code:

T = Table of coefficients required.
G = Graphical method of solution.
WE = Only working equation needed.
C = Curve comparison method available.
I = Independent calculation of each class interval.
S = Sequential calculations required.
L = Logarithmic scale.

TABLE 6. LIMITS FOR GROUPED PLANAR SECTIONS FROM SPHERES

Group No.	Relative Section Diameter, d/d_{max}	Relative Section Areas, A/A_{max}
1	1.0000	1.0000 - 0.6310
2	0.7943	0.6310 - 0.3981
3	0.6310	0.3981 - 0.2512
4	0.5012	0.2512 - 0.1585
5	0.3981	0.1585 - 0.1000
6	0.3162	0.1000 - 0.0631
7	0.2512	0.0631 - 0.0398
8	0.1995	0.0398 - 0.0251
9	0.1581	0.0251 - 0.0158
10	0.1259	0.0158 - 0.0100
11	0.1000	0.0100 - 0.0063
12	0.0794	0.0063 - 0.0040

in the brackets, i.e., for i = 5, i-1 = 4, i-2 = 3, i-3 = 2, and i-4 = 1.

To show how the calculations are made, let us determine $(N_V)_4$ from the data given in Table 7. The equation obtained in this case for j = 4 (= i) is

$$(N_V)_4 = \frac{1}{D_4}\left[1.6461(N_A)_4 - 0.4561(N_A)_3 - 0.1162(N_A)_2 - 0.0415(N_A)_1\right] \quad (20)$$

Substituting the experimental data, we obtain

$$(N_V)_4 = \left[\frac{1}{0.0316} \quad 1.65(230) - 0.456(253) - 0.116(161) - 0.0415(104)\right] =$$

$$7630 \text{ mm}^{-3}. \quad (21)$$

If the complete distribution curve is desired, this type of calculation would be performed for all particle sizes. The total of all $(N_V)_j$ gives N_V, the total number of particles per unit volume. The complete results from the calculations are summarized in Table 8. It can be seen that this method for calculating N_V is direct, simple, and eminently useful. Of further interest is the possibility of analyzing systems of convex particles of more complex shape.

We may wish to express the size distribution in terms of the three numerical parameters, $\overline{D}$, $\sigma(D)$, and N_V, instead of the size distribution curve itself. These values are obtained readily. The mean diameter is expressed by

$$\overline{D} = \frac{1}{N_V}\sum_{j=1}^{j=k} (N_V)_j D_j, \quad (22)$$

the standard deviation by

$$\sigma(d) = \left[D_j^2 - \overline{D}^2\right]^{1/2} \quad (23)$$

and the total number of particles per unit volume by

$$N_V = \sum_{j=1}^{j=k} (N_V)_j , \quad (24)$$

*For example, K. & E., 2-cycle, No. 359-24.

TABLE 7. MEASURED DISTRIBUTION OF FERRITE GRAIN SECTION SIZES

Class Interval	Range of Section Diameters, d_i, mm	Relative Section Areas, A/A_{max}	No. of Sections per mm^2, $(N_A)_i$
1	0.0631 - 0.0501	1.0000 - 0.6310	104
2	0.0501 - 0.0398	0.6310 - 0.3981	161
3	0.0398 - 0.0316	0.3981 - 0.2512	253
4	0.0316 - 0.0251	0.2512 - 0.1585	230
5	0.0251 - 0.0199	0.1585 - 0.1000	128
6	0.0199 - 0.0158	0.1000 - 0.0631	69
			N_A = 955 mm^{-2}

TABLE 8. CALCULATED DISTRIBUTION OF FERRITE GRAIN SIZES

Class Interval	Diameter of Particles, D_j, mm	No. of Grains per mm^3, $(N_V)_j$
1	0.0631	2,713
2	0.0501	4,341
3	0.0398	8,313
4	0.0316	7,630
5	0.0251	3,359
6	0.0199	491
		N_V = 26,847 mm^{-3}

where k is the number of class intervals. Thus, from Table 8 we obtain $\bar{D}$ = 0.0393 mm, $\sigma(D)$ = 0.012 mm, and N_V = 26,847 mm^{-3}. An alternative graphical method is to plot the cumulative percentages of $(N_V)_j$ versus particle diameter on log-probability graph paper*. If the size distribution conforms to the log-normal distribution, as most particle- and grain-size distributions do, a straight line will ensue. Then the values of $\bar{D}$ and $\sigma(D)$ can be read directly from the curve: $\bar{D}$ at a cumulative frequency of 50, and $\sigma(D)$ between either 84.13 and 50, or 50 and 15.87.

Another method for obtaining a particle size distribution is by measurement of the intercept chord length distribution. (16) From the standpoint of ease of gathering data, the chord methods are quite promising, especially since the advent of electronic scanning devices. Bockstiegel gives an improved derivation of the chord intercept relationship for spheres (15). The number of chords per unit length $(n_L)_i$, $(n_L)_{i+1}$, etc., are obtained experimentally and grouped into suitable class intervals, ℓ_{i-1} to ℓ_i, ℓ_i to ℓ_{i+1}, etc., respectively. We wish to obtain $(N_V)_{i-1/2}$, which represents the number of particles per unit volume with diameters between $\ell_{i-1/2}$ and $\ell_{i+1/2}$. His general equation is

$$(N_V)_{i+1/2} = \frac{4}{\pi}\left[\frac{(n_L)_i}{\ell_i^2 - \ell_{i-1}^2} - \frac{(n_L)_{i+1}}{\ell_{i+1}^2 - \ell_i^2}\right], \qquad (25)$$

which is valid for any kind of class interval divisions. Note that N_V can be obtained independently for any size group, and that tables of coefficients are not required.

A further simplification of Equation (25) is possible by defining logarithmic class intervals such that $\ell_{i+1} = z\ell_i$. Putting $z = \sqrt{2}$ gives $\ell_{i+1}^2 = 2\ell_i^2$, $\ell_i^2 = 2\ell_{i-1}^2$, etc., which when inserted into Equation 25 gives

$$(N_V)_{i+1/2} = C\,\frac{2(n_L)_i - (n_L)_{i-1}}{(2)^i}, \qquad (26)$$

where $C = 4/\pi\ell_o^2$ is a constant independent of i, and ℓ_o is the upper limit of the lowest class interval. If relative values, $(N_V)_{i+1/2}/\Sigma(N_V)_{i+1/2}$, are desired rather than the absolute numbers, $(N_V)_{i+1/2}$, the constant C cancels out. Thus the relative size distribution is obtained directly from the experimental data.

As an example of the application of Equation (25), consider the case for i = 4 if we have the following data:

Group No.	Range of Chord Lengths, mm	No. of Chords per mm, $(N_L)_i$	Diameter of Particles, mm
4	0.0075-0.0100	19	0.0100
5	0.0100-0.0125	13	0.0125

Substitution in Equation (25) gives

$$(N_V)_{i+1/2} = \frac{4}{\pi}\left[\frac{19}{(10^2-7.5^2)10^{-6}} - \frac{13}{(12.5^2-10^2)10^{-6}}\right] =$$

$$\frac{4}{\pi}\left[\left(\frac{19}{43.75} - \frac{13}{56.25}\right)10^6\right] = 259{,}000 \text{ mm}^{-3}.$$

Calculation at i = 4 according to Equation (26) results in

$$(N_V)_{i+1/2} = C\left(\frac{2 \times 19-13}{(2)^4}\right) = C\left(\frac{25}{16}\right) = C\ (1.56).$$

This result would be divided by $\sum_{i=1}^{i=k}(N_V)_{i+1/2}$ in order to get the relative particle frequency at i=4. Occasionally, negative values are obtained for the smallest particles, but this may happen using any method. (See the discussion of this problem in Reference 1.) A practical solution if this occurs is to equate the negative values to zero.

PROJECTED IMAGES

In general, microscopists encounter two main types of projected images. In one type, the image results from a transmitted beam through the specimen, giving a representation of the features located within the three-dimensional space. In the other main type, the projected image is generated by a reflected beam from the external surface of the specimen.

At present, only the most rudimentary calculations can be made on images projected by the reflection techniques (17,18), and we will not dwell upon this aspect of the problem. Treatment of images of the first type, however, have matured to a considerably greater extent (19,20) and the basic relationships will be reviewed here.

Figure 4 reveals pictorially some fundamental relationships between microstructural features in a plane or a foil and their corresponding images projected to a line or surface. Projected quantities are identified by means of a prime.

There are three basic equations that underlie most of our relationships for projected quantities. The first applies to a straight linear segment in a plane projected over all orientations to a projection line, and is

$$L_2 = (\pi/2)\overline{L}', \tag{27}$$

where $\overline{L}'$ is the mean projected length on the projection line.

The second basic relationship pertains to a linear segment of a curve of any curvature in space, projected over all orientations to a projection plane. The result is

$$L_3 = (4/\pi)\overline{L}', \tag{28}$$

where $\overline{L}'$ is the mean projected length on the projection plane.

The third basic equation is

$$S = r\,\overline{A}', \tag{29}$$

which is the Cauchy equation for the surface area of a convex body, S. The quantity $\overline{A}'$ is the mean projected area of the convex body, projected over all orientations to a projection plane.

Corresponding equations can be written for systems of lines or surfaces. They are

$$L_A = (\pi/2)L_L'/h \tag{30}$$

for a random system of lines in a plane of height h, with negligible overlap, where L_L' is the projected line length per unit length of test line along the projection axis.

$$L_V = (4/\pi)L_A'/t \tag{31}$$

applies to a random system of lines in a foil of thickness t, and L_A' is the projected line length per unit area in the projection plane. This is the equation used most in studies of dislocations by means of thin foil transmission electron microscopy.

The Cauchy relationship for a system of randomly oriented convex particles with negligible overlap is

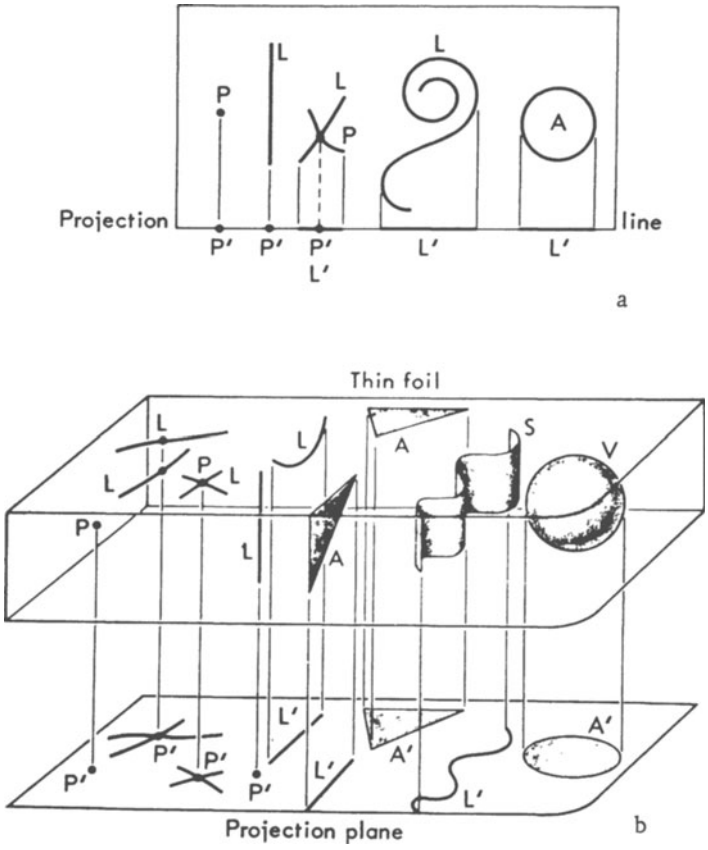


Figure 4. Projection of (a) figures in a plane onto a projection line, and (b) structural features in a foil onto a projection plane.

$$S_V = A_A'/t \tag{32}$$

where A_A' is the projected area per unit area in the projection plane ($= P_P'$), and t is the foil thickness.

The dislocation density is usually best expressed in terms of L_V and not P_A (see Equation (31), since L_V represents a unique value for any system of lines, regardless of configuration, when random sampling is employed. P_A, on the other hand, can vary depending on the degree of orientation of the lines and on the angle of the test plane with respect to the orientation axis. In some cases, of course, special methods are advantageous that use non-random measurements of P_A, but then, other equations must be employed [see, for example, Equation (6)].

Projected images of dislocation lines can be determined in two ways -- by an intersection count P_L' on the projection plane using

$$L_V = 2P_L'/ \tag{33}$$

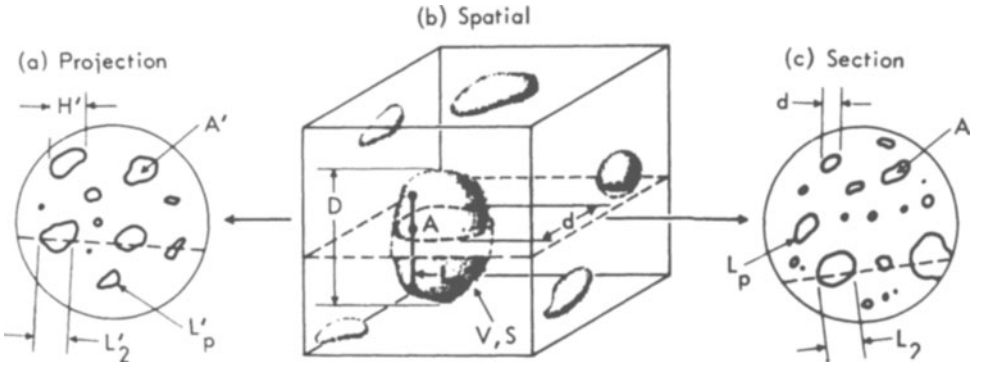
or by an areal point count P_A' of the terminal points of dislocation lines that cut the foil surfaces, by means of

$$L_V = P_A' \ . \tag{34}$$

Both methods are useful depending on the dislocation configurations and experimental techniques available.

In Figure 5 we provide an interrelated and interconsistent set of equations for convex particles, their projections, and their sections. (1,21) The quantities are observed either on the plane of polish or in the projected image. These relationships are general in nature and free of assumptions, with the proviso that particle overlap and truncation are negligible. Note that L_2', L_3, and L_2 are intercept lengths; H', D and d are tangent diameters; L_p' and L_p are perimeter lengths; A' is the projected area and A the intercept area. Frequently, the simple relationships shown in Figure 5 may be used as approximations for real particles.

Quantities of interest in a study of the properties of individual particles are the mean particle volume $\overline{V}$, the mean surface area $\overline{S}$, the mean projected area $\overline{A}'$, the mean projected height $\overline{H}'$, the mean intercept area $\overline{A}$, and the mean intercept length $\overline{L}_3$. The more important parameters used to characterize a system of convex particles in space are the volume fraction V_V, the number per unit volume N_V, the number of particle interceptions per unit area of



$$\bar{A}' = \bar{L}'_2 \cdot \bar{H}'$$
$$\bar{L}'_2 = \pi\bar{A}'/\bar{L}'_p$$
$$\bar{H}' = \bar{L}'_p/\pi$$
$$N'_L = N'_A \cdot \bar{H}'$$
$$\bar{L}'_2 = \bar{d}$$
$$\bar{H}' = \bar{A}'/\bar{d}$$

$$V = \bar{A} \cdot \bar{D} = \bar{A}' \cdot \bar{L}_3$$
$$S = 4\bar{A}'$$
$$\bar{A} = \bar{L}_3 \cdot \bar{L}'_2$$
$$\bar{L}_3 = 4V/S$$
$$\bar{D} = \bar{H}'$$
$$\bar{L}_3 = \bar{L}_2$$

$$\bar{A} = \bar{L}_2 \cdot \bar{d}$$
$$\bar{L}_2 = \pi\bar{A}/\bar{L}_p$$
$$\bar{d} = \bar{L}_p/\pi$$
$$N_L = N_A \cdot \bar{d}$$
$$N_A = N_V \cdot \bar{D}$$
$$N_V = N_L/\bar{A}'$$

Figure 5. Relationships among convex particles in space, their sections and their projectors.

the test plane N_L, the number of particle interceptions per unit length of the test line N_L, and the mean free distance λ. The quantities enumerated above have been collected and summarized in Table 9 for the sphere, truncated octahedron (or tetrakaidecahedron) and for convex particles in general. Tabulations of this type (22) afford the microscopist the opportunity to approximate his microstructures with particles of known shapes.

The previous paragraphs dealt with the properties and equations of convex particles and systems of particles under simplified conditions. Exact equations can be written if the particle shape is known, provided there is no truncation of particles by foil surfaces, no image overlap on the projection plane, and no reentrant surfaces on the particles. Under actual conditions, however, these factors inevitably come into play, and whether they can be neglected depends primarily on the accuracy required, the type of microstructure, and the experimental limitations.

Approximate corrections can be made for truncation and overlap effects, under favorable conditions, when the volume fraction of particles is small and when the ratio of foil thickness to mean particle diameter does not exceed 2 or 3*. We give first the basic equation for volume fraction of convex particles in the absence of truncation and overlap

$$V_V = A_A' \, \bar{L}_3/t \tag{35}$$

which, for spherical particles of mean diameter $\bar{D}$ under the same conditions, becomes

$$V_V = 2A_A' \, \bar{D}/3t \tag{35a}$$

or, in terms that are measurable on the projection plane,

$$V_V = 2 \, P_p' N_L/3N_A' t \; . \tag{35b}$$

Note that the foil thickness t must be determined separately and that $\bar{D}_L' = N_L'/N_A'$.

When truncation is considered, we have for convex particles

$$V_V = A_A' - S_V t/4, \tag{36}$$

and for spherical particles,

$$V_V = A_A' \left(\frac{2D}{2D+3t} \right) . \tag{36a}$$

*A correction for overlap of spheres should be unnecessary if $V_V(t/\bar{D}) < 0.04$.

TABLE 9. PROPERTIES OF A SPHERE, TRUNCATED OCTAHEDRON AND CONVEX PARTICLES IN GENERAL

Property	Sphere, D = 2r	Truncated Octahedron, Edge Length = a	General Equations for Convex Particles
V	$4\pi r^3/3$	$11.314\ a^3$	$\bar{V}=\bar{A}'\bar{L}_3=\bar{A}\ \bar{H}'$
S	$4\pi r^2$	$26.785\ a^2$	$\bar{S}=4\bar{A}'=4\ \bar{V}/\bar{L}_3$
$\bar{A}'$	πr^2	$6.696\ a^2$	$\bar{A}'=\bar{S}/4=\bar{V}/\bar{L}_3$
$\bar{H}'$	$2r$	$3.0\ a$	$\bar{H}'=\bar{V}/\bar{A}=\bar{A}'\bar{L}_3/\bar{A}$
$\bar{A}$	$2\pi r^2/3$	$3.77\ a^2$	$\bar{A}=\bar{V}/\bar{H}'=\bar{A}'\bar{L}_3/\bar{H}'$
$\bar{L}_3$	$4r/3$	$1.69\ a$	$\bar{L}_3=4\bar{V}/\bar{S}=\bar{A}\ N_A/N_L$
r,a,ρ	$4=2N_L/\pi N_A$	$a=0.45N_L/N_A$	$\rho=\bar{H}'/2=N_A\bar{A}'/2N_L=\bar{A}'/2\bar{L}_2'$
N_V	$\pi N_A^2/4N_L$	$0.744N_A^2/NL$	$N_V=N_A/\bar{H}'=N_L/\bar{A}'$

$$V_V=N_V\ \bar{V}=N_A\bar{A}=N_L\ \bar{L}_3$$

When $t \to \infty$, $V_V \to A_A'$; and for $t \gg D$ truncation becomes relatively less important and Equation (35a) is obtained. S_V must either be determined separately or approximated by assuming a particle shape.

When the correction for overlap is included with that for truncation, we obtain for convex particles

$$V_V = -\ln\,(1-A_A') - S_V t/4\ , \tag{37}$$

and for spherical particles

$$V_V = -\ln\,(1-A_A')\left(\frac{2D}{(2D + 3t}\right)\ . \tag{37a}$$

This overlap correction allows only for the shadowing of one particle by another, and not the self-overlap caused by reentrant surfaces on a particle. In order to correct for self overlap, we must resort to the concept of "total projection" (19) but this topic would take us too far afield.

We close this brief treatment of projected images with the equations needed for a three-parameter representation of the size distributions of convex particles or spheres (23). They are:

(1) The mean projected particle diameter $\overline{H}'$ for convex particles

$$\overline{H}' = N_L'/N_A'\ , \tag{38}$$

(2) The number of particles per unit volume

$$N_V = N_A'/t, \tag{39}$$

which is valid in the absence of overlap and truncation, or

$$N_V = N_A'/(t+\overline{H}'), \tag{39a}$$

which accounts for truncation only. Overlap would affect N_A', and approximate corrections are available in the literature.

(3) The standard deviation

$$\sigma(H') = \left[\overline{(H')2/j} - \overline{(H')2/j}\right]^{1/2} \tag{40}$$

which, for spheres of mean diameter $\overline{D}$, can be written

$$[\sigma(D)]^2 = \frac{-4\ln(1-A_A')}{\pi N_A'} - (\overline{D})^2 \tag{40a}$$

This equation corrects for truncation and overlap in systems of small volume fraction.

There are many other aspects of the subject of Quantitative Stereology not covered in this article. However, with the steadily increasing availability of good articles and texts, the microscopist should experience no difficulty in applying these, and more advanced, quantitative methods to his work.

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(23) Reference 1.